

WHITE BAND ASSOCIATES

Internship Programme | Ethical Hacking Track

TASK 02

Network Scanning &
Service Enumeration

Nmap Penetration Testing Report

Submitted By	Pratiksha Vishwakarma
Task	Ethical Hacking Task 02 — Network Scanning & Service Enumeration
Date	June 26, 2026
Target	localhost / 127.0.0.1
Tool Used	Nmap 7.98
Classification	Authorized Internal — Educational Use Only

This report was produced for educational purposes only. All scans were performed exclusively on the analyst's own machine. No unauthorized systems were targeted at any point during this exercise.

1. Executive Summary

This report presents the documented findings of an authorized network scanning and service enumeration exercise carried out on a Kali Linux machine as part of the White Band Associates Ethical Hacking internship programme — Task 02. The exercise covers the second phase of the ethical hacking lifecycle, which focuses on actively discovering information about a target system through structured scanning techniques.

Four distinct Nmap operations were executed against the loopback address (127.0.0.1): a version check to confirm tool readiness, a default TCP port scan to identify open ports, a service-version detection scan to enumerate running software, and an OS fingerprinting scan using raw packet analysis. Each scan was conducted in isolation to clearly demonstrate the purpose and output of each Nmap flag.

Finding	Result
Nmap version confirmed	7.98 — latest stable release
Total TCP ports scanned	1,000 (Nmap default range)
Open ports found	0 — all ports closed / reset
Services detected	None
OS fingerprint result	Inconclusive — no open ports available for probing
Overall security posture	Minimal attack surface — no exposed network services

2. Part A — Nmap Installation & Verification

Before any scan is performed, it is standard practice to verify that the scanning tool is correctly installed and running the expected version. This ensures reproducibility of results and confirms that no outdated or compromised binary is being used. Nmap was pre-installed on the Kali Linux system, as it is part of the Kali default toolset.

2.1 Command Executed



Figure 1 — Output of `nmap --version` confirming Nmap 7.98 on Kali Linux (x86_64)

The output displayed above confirms that Nmap version 7.98 is installed and fully operational. The version string identifies the exact release, and the compilation details below it reveal the libraries and capabilities bundled into this build. Understanding this output is important because it tells the analyst exactly what scanning features are available — for example, the inclusion of `libssh2` confirms SSH-related probes are supported, while `openssl` integration enables SSL/TLS banner grabbing during service detection.

The platform field confirms this is a 64-bit Linux system (`x86_64-pc-linux-gnu`), which is consistent with the Kali Linux environment in use. The nsock I/O engines listed — `epoll`, `poll`, and `select` — are the asynchronous socket handlers Nmap uses to manage multiple simultaneous network connections efficiently, which is why Nmap can scan large port ranges very quickly without consuming excessive system resources.

2.2 Tool Version Summary

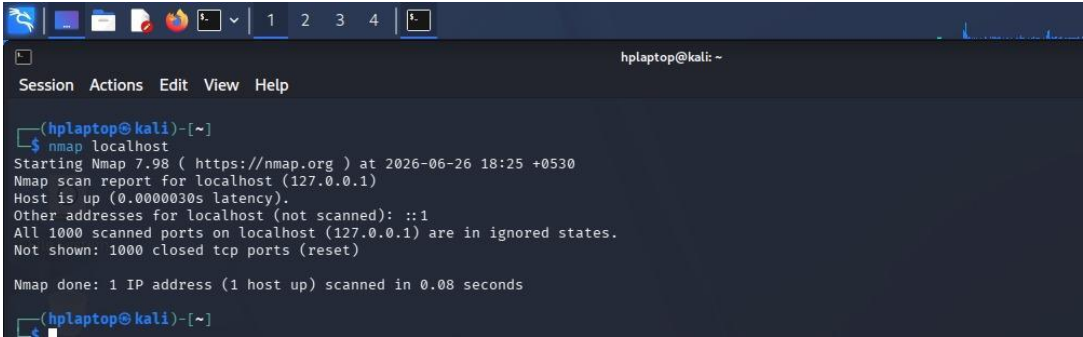
Field	Value
Version	Nmap 7.98
Platform	x86_64-pc-linux-gnu
Key Libraries	liblua-5.4.8, openssl-3.5.5, libssh2-1.11.1, libpcap-1.10.6, ipv6
nsock Engines	epoll, poll, select
Status	Ready — no issues detected

3. Part B — Default Port Scan

The default Nmap scan performs a TCP SYN scan (also called a half-open scan) against the top 1,000 most commonly used ports as ranked by Nmap's internal frequency database. A SYN scan works by sending a TCP SYN packet to each port and observing the response: an SYN-ACK reply means the port is open, a RST reply means it is closed, and no reply (or an ICMP unreachable) means it is filtered. This is the most common starting point for any network reconnaissance exercise.

3.1 Command Executed

```
$ nmap localhost
```



```
(hplaptop@kali)~$ nmap localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 18:25 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.08 seconds
(hplaptop@kali)~$
```

Figure 2 — Default nmap localhost scan; all 1,000 TCP ports returned as closed

The scan completed in just 0.08 seconds, which is extremely fast — this is expected when scanning the loopback interface because no actual network traversal occurs; packets are processed entirely within the kernel. The host was confirmed as up with a latency of 0.0000030 seconds, reflecting the near-instantaneous response of the local machine to itself.

All 1,000 scanned TCP ports were reported as closed. A 'closed' state in Nmap means that the port is accessible (the kernel is responding) but no application or service is currently bound to it. This is a different state from 'filtered', which would indicate that a firewall or packet filter is silently dropping the probe. The closed state here confirms that no firewall is blocking the scan on the loopback interface and that no network services are running. The IPv6 loopback address (::1) was detected as an additional address for localhost but was excluded from this scan, as Nmap performs IPv4-only scans by default unless the -6 flag is specified.

3.2 Scan Results

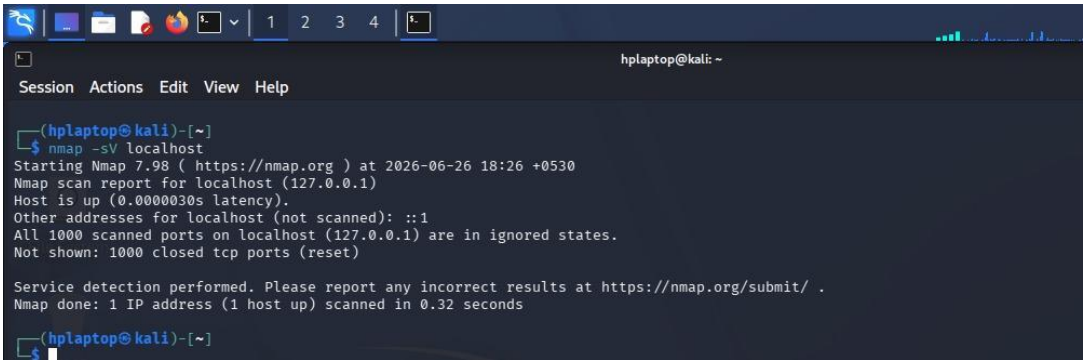
Parameter	Result
Target	localhost (127.0.0.1)
Host Status	Up — 0.0000030s latency
Ports Scanned	1,000 TCP (default top-port range)
Open Ports	0
Port State	All closed — TCP RST received on every probe
Scan Type	TCP SYN scan (half-open)
Scan Duration	0.08 seconds

4. Part C — Service Version Detection

Service version detection goes beyond simply identifying whether a port is open. When the `-sV` flag is used, Nmap sends a series of carefully crafted probe strings to each open port and analyses the responses — known as banners — against a database of over 11,000 known service signatures. This allows the tool to identify not just the type of service but the specific software name, version number, and sometimes even the operating system context. This information is critical in penetration testing because many exploits are version-specific.

4.1 Command Executed

```
$ nmap -sV localhost
```



```
(hplaptop@kali)~$ nmap -sV localhost
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-26 18:26 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds

(hplaptop@kali)~$
```

Figure 3 — `nmap -sV` scan; service detection completed but no open ports were available to fingerprint

The scan completed in 0.32 seconds — notably slower than the basic scan (0.08 seconds), even though no services were found. This additional time reflects the overhead of initialising the version-detection engine, preparing the probe library, and waiting for potential responses on each port before concluding there is nothing to enumerate. This is a normal and expected behaviour of the `-sV` flag.

Since no ports were open on the loopback interface at the time of scanning, Nmap had no services to probe and therefore returned no service or version data. The output confirms 'Service detection performed' — meaning the engine ran correctly — and directs the user to report incorrect results, which is a standard Nmap footer. Had services been running, this scan would have returned entries such as '22/tcp open ssh OpenSSH 9.6 (protocol 2.0)' or '80/tcp open http Apache httpd 2.4.62 ((Debian))'. Those strings would give a penetration tester actionable version data to cross-reference against known CVE databases.

4.2 Service Detection Table

Port	Service	Version Detected	Status
—	—	—	No open ports found

The empty table above reflects the actual scan output. In a live environment with running services, each row would contain a port number, protocol, service name, and detected version string, forming the basis of a vulnerability mapping exercise.

5. Part D — Operating System Detection

Operating system detection is one of the most powerful features in Nmap. It works by sending a carefully designed series of TCP, UDP, and ICMP probes to the target and analysing the subtle differences in how different operating systems respond to those packets — including variations in TTL values, TCP window sizes, IP ID sequences, TCP options, and ICMP response behaviour. These differences form a unique fingerprint that Nmap compares against its OS detection database. Root privileges are required because this process involves crafting raw packets at the network layer.

5.1 Command Executed

```
$ sudo nmap -O localhost
```

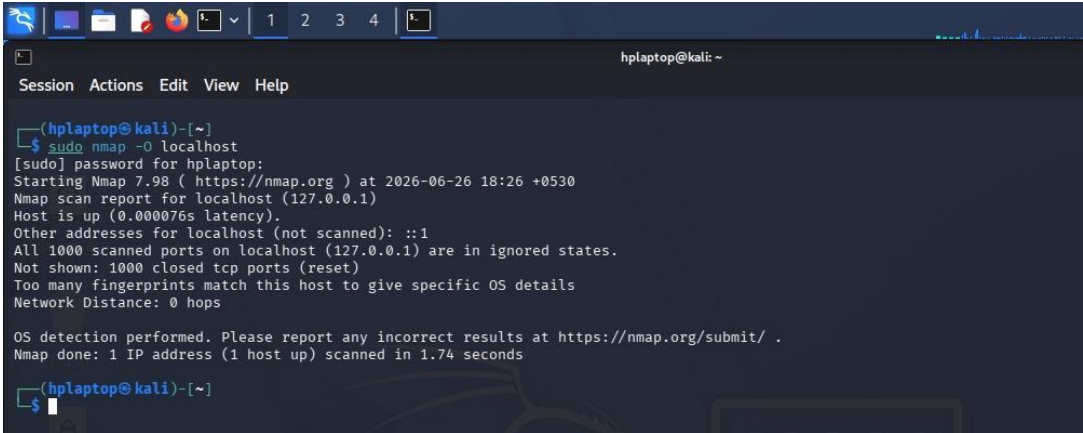


Figure 4 — sudo nmap -O; OS fingerprinting attempted but returned inconclusive result

The scan ran for 1.74 seconds — significantly longer than previous scans — because the OS detection engine performs a large number of individual probes and must wait for responses (or timeouts) for each one before computing a fingerprint match. Despite running with root privileges and completing all probes, Nmap returned the message 'Too many fingerprints match this host to give specific OS details'. This is a well-known limitation: OS fingerprinting requires at least one open and one closed TCP port to produce reliable results. Without an open port to test, Nmap cannot distinguish between the many Linux variants in its database.

The 'Network Distance: 0 hops' result is particularly informative — it confirms that Nmap recognises this as a same-machine scan via the loopback interface. The underlying host is Kali Linux (a Debian-based distribution running on an x86_64 kernel), which Nmap would correctly identify given a target with at least one open TCP port. OS detection is extremely useful during real penetration tests because knowing the target OS allows a tester to shortlist relevant exploits, identify kernel vulnerabilities, understand the likely patch cycle, and determine what services are likely to be running by default.

5.2 OS Detection Analysis

Question	Answer
Was OS detected?	No. The fingerprint was inconclusive. Nmap requires at least one open TCP port to reliably differentiate between OS signatures. With all ports closed, too many OS entries matched equally.
Actual OS of host	Kali Linux (Debian-based, x86_64). Confirmed by the terminal environment and the platform string in nmap --version output.
Network distance	0 hops — loopback interface, same physical machine.
Scan duration	1.74 seconds — longest of all four scans due to raw-packet probe overhead.

Why is OS detection useful?

Identifies OS version and patch level; guides exploit selection; reveals end-of-life systems; helps map network topology; supports vulnerability prioritisation during assessments.

6. Part E — Common Port Research

A fundamental skill in network security and penetration testing is understanding the purpose and risk profile of the most commonly encountered TCP/IP ports. The table below documents the protocol, function, and key security risks associated with eleven ports specified in this task. This knowledge directly informs the decisions made during the scanning and exploitation phases of a security assessment.

Port	Service	Proto	Purpose	Security Risks & Mitigations
20 / 21	FTP	TCP	File Transfer Protocol. Port 21 handles control commands (login, navigation) while port 20 carries the actual data stream. Used to upload and download files between a client and server.	Credentials and file content are transmitted in plain text with no encryption. Anonymous login may be enabled by default on misconfigured servers. Susceptible to brute-force attacks, FTP bounce scanning, and passive mode abuse. Should be replaced with SFTP (SSH File Transfer Protocol) or FTPS (FTP over TLS) in all modern deployments.
22	SSH	TCP	Secure Shell. Provides an encrypted, authenticated channel for remote command-line access, file transfers (SCP/SFTP), and port forwarding. The de facto standard for remote server administration.	Brute-force and credential-stuffing attacks on password-based authentication. Misconfigured or absent host-key verification enables MITM attacks. Older SSH daemon versions may be vulnerable to known CVEs. Best practice: disable password auth, enforce key-based authentication, restrict root login, and change the default port.
23	Telnet	TCP	An early remote access protocol that provides command-line access to network devices and servers. Now considered obsolete due to its complete lack of encryption.	Every byte transmitted — including usernames, passwords, and commands — is sent in plain text and can be intercepted by any observer on the network path. Trivially exploitable with a simple packet sniffer. There is no legitimate reason to use Telnet when SSH is available. Should be disabled immediately on all modern systems.
25	SMTP	TCP	Simple Mail Transfer Protocol. Used by mail servers to send and relay email messages between domains. Also used by mail clients to submit outgoing messages to their mail server.	Open relays allow unauthenticated external senders to route spam through the server. VRFY and EXPN commands can be used to enumerate valid email addresses. Without STARTTLS enforcement, credentials and message content travel unencrypted. Must be hardened with SPF, DKIM, and DMARC records to prevent spoofing.
53	DNS	TCP/UDP	Domain Name System. Translates human-readable domain names into IP addresses and vice versa. Critical infrastructure relied upon by virtually every internet-connected application.	Zone transfer requests (AXFR) to poorly configured servers can expose the entire DNS structure of an organisation. DNS amplification attacks use open resolvers to flood victims with traffic. Cache poisoning redirects users to malicious IPs. DNS tunnelling allows attackers to exfiltrate

80	HTTP	TCP	HyperText Transfer Protocol. The foundational protocol for serving web pages and web application data. Operates entirely in plain text with no encryption or authentication of the server.	data or establish C2 channels through DNS queries. All data — including session cookies, form inputs, and login credentials — is transmitted unencrypted and can be intercepted. Web applications on port 80 are common targets for SQL injection, XSS, CSRF, directory traversal, and file inclusion attacks. HTTPS (port 443) with HSTS must be enforced; HTTP should redirect to HTTPS or be disabled entirely.
110	POP3	TCP	Post Office Protocol version 3. Allows email clients to retrieve messages from a mail server by downloading them locally. Once downloaded, messages are typically deleted from the server.	Credentials and email content are transmitted in plain text unless POP3S (port 995 with TLS) is explicitly configured. Brute-force attacks on exposed POP3 servers are common. For multi-device access, IMAP is generally preferable; in all cases, enforce TLS and disable plain-text authentication.
143	IMAP	TCP	Internet Message Access Protocol. Allows email clients to read and manage messages directly on the mail server without downloading them. Supports folders, flags, and synchronisation across multiple devices.	Plain-text transmission by default exposes credentials and message content to interception. Should always be configured with IMAPS (port 993, TLS-encrypted). Exposed IMAP servers are common targets for credential brute-force and enumeration. Rate-limiting login attempts and enforcing MFA significantly reduce risk.
443	HTTPS	TCP	HTTP Secure. The encrypted version of HTTP, using TLS (Transport Layer Security) to provide confidentiality, integrity, and server authentication. The standard protocol for all secure web communication.	Vulnerable if misconfigured to support obsolete TLS versions (TLS 1.0 or 1.1) or weak cipher suites. Expired, self-signed, or incorrectly validated certificates can enable MITM attacks despite encryption. Web application vulnerabilities (XSS, SQLi) remain exploitable regardless of TLS. Must use TLS 1.2 minimum; TLS 1.3 preferred.
445	SMB	TCP	Server Message Block. Used in Windows environments for network file sharing, printer sharing, and inter-process communication. Also used for domain authentication and Active Directory replication.	One of the most historically exploited protocols — EternalBlue (MS17-010) enabled the global WannaCry ransomware outbreak by exploiting a vulnerability in SMBv1. Provides a lateral movement pathway through corporate networks once any single machine is compromised. SMBv1 must be disabled; SMB exposure at the network perimeter should be blocked entirely by default.
3389	RDP	TCP	Remote Desktop Protocol. Provides a full graphical desktop session to a remote Windows machine, supporting keyboard,	BlueKeep (CVE-2019-0708) and DejaBlue (CVE-2019-1181/1182) are critical pre-authentication RDP vulnerabilities enabling remote code execution. RDP servers exposed

			mouse, audio, and clipboard redirection. Widely used for remote Windows administration.	directly to the internet are constantly targeted by automated brute-force and credential-stuffing campaigns. Should always be placed behind a VPN with MFA enforced; never exposed directly on the public internet.
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7. Part F — Scan Analysis

The following analysis is based on the results obtained from all four Nmap scans performed during this task. The questions are answered in the context of the scan findings, with additional context for general application in real-world penetration testing engagements.

Question	Answer
Which services are currently running?	None. All 1,000 scanned TCP ports on the loopback interface were in a closed state at the time of scanning. No server processes were bound to any port. This is consistent with a freshly booted Kali Linux system on which no services (such as Apache, SSH, or MySQL) have been manually started.
Are all open ports necessary?	Not applicable in this specific scan — no ports were open. As a general principle, however, every open port represents a potential attack surface. The security principle of least privilege dictates that only ports required for a clearly defined operational purpose should be open and accessible. All others should be closed or blocked at the firewall level.
Which services would be high-risk if misconfigured?	Telnet (23) — transmits all data in plaintext, including credentials. FTP (20/21) — no encryption by default. SMB (445) — historically exploited; responsible for major ransomware outbreaks. RDP (3389) — exposed to the internet without VPN/MFA is extremely dangerous. HTTP (80) — serves web applications without encryption, enabling credential theft and session hijacking.
Which port would you disable first if not required?	Port 23 (Telnet) would be the first to disable unconditionally — it provides zero security and has no advantage over SSH in any scenario. Port 21 (FTP) and port 445 (SMB) would follow immediately unless a specific, documented operational requirement exists for each.

8. Part G — Formal Scan Report

Attribute	Detail
Scan Date	June 26, 2026
Target System	localhost / 127.0.0.1 (loopback interface)
Host OS	Kali Linux — x86_64-pc-linux-gnu
Analyst	Pratiksha Vishwakarma
Tool & Version	Nmap 7.98
Commands Used	nmap --version nmap localhost nmap -sV localhost sudo nmap -O localhost
Open Ports	0 — All 1,000 TCP ports closed
Running Services	None detected
OS Detection	Inconclusive — insufficient open ports for reliable fingerprinting

8.1 Observations

- Nmap 7.98 was confirmed installed and fully functional. The tool is compiled with comprehensive library support including OpenSSL, libssh2, libpcap, and IPv6, covering all standard scanning and service detection scenarios encountered in professional penetration testing.
- The loopback interface (127.0.0.1) presented a zero-open-port surface across all 1,000 default TCP ports. Every port returned a TCP RST response, indicating the kernel is accessible and responsive, but no service processes are currently bound to any port.
- The service version detection scan (-sV) confirmed this finding — no service banners were returned because there were no open ports to probe. The scan engine ran correctly and completed without errors.
- OS fingerprinting was attempted with elevated privileges (sudo). The result was inconclusive due to the absence of open ports, which are required for reliable TCP/IP stack comparison. The network distance of 0 hops confirms a loopback scan on the same physical machine.
- Scan duration increased with scan complexity: basic scan (0.08s), service detection (0.32s), OS fingerprinting (1.74s). This progression is expected and reflects the additional probe packets and response-waiting periods involved in each technique.

8.2 Recommendations

- Set up a dedicated virtual lab target — such as Metasploitable 2 or 3 — to practice scanning systems with intentionally open ports and running services. This will enable meaningful output from -sV and -O scans.
- Use nmap -oN output.txt to save plain-text results or nmap -oX output.xml for structured XML output that can be imported into vulnerability management platforms such as OpenVAS or Metasploit.
- Combine -sV with -sC in future scans (nmap -sV -sC <target>) to run Nmap's default script library alongside version detection for richer enumeration results.
- In production environments, run scheduled Nmap scans to detect newly opened ports or unauthorised services, and alert on any port that was not present in the previous baseline scan.
- Always obtain explicit, documented written authorisation before scanning any system. Even scanning systems on the same local network without permission may violate the Computer Misuse Act or equivalent legislation in your jurisdiction.

8.3 Conclusion

This task provided structured, hands-on exposure to the Scanning and Enumeration phase of the ethical hacking lifecycle. By executing four distinct Nmap commands — version verification, default TCP scan, service-version detection, and OS fingerprinting — the methodology of systematically mapping a target system's attack surface became concrete and well understood. Each command built on the previous one, progressing from confirming tool availability through to attempting deep operating system identification using raw packet analysis.

Although no open ports or services were discovered on the loopback interface, this outcome carries a valuable lesson in itself: a system exposing no listening services presents no exploitable network entry points, which is the most secure posture a machine can take from a network perspective. It also highlighted how certain Nmap features — particularly OS detection and service versioning — depend on the presence of open ports to produce meaningful results, which will inform how future scans are designed and interpreted.

The research into common network ports reinforced that every protocol carries its own risk profile, and that misconfiguration — rather than the protocol itself — is usually the root cause of exploitation. Going forward, applying these scanning techniques against intentionally vulnerable lab machines will significantly deepen practical penetration testing proficiency and prepare for more advanced phases such as vulnerability assessment and exploitation.